Fiscal Year:	FY 2010	**Task Last Updated:**	FY 05/25/2010
PI Name:	Kanas, Nick M.D.		
Project Title:	Effects of High vs. Low Autonomy on Space Crewmember Performance (Autonomy)		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94121	**Congressional District:**	8
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	08/12/2005	**End Date:**	03/31/2010
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Shea, Cami1e	**Contact Phone:**	281-244-2017
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Flight Program:			
Flight Assignment:	NOTE: End date should be 3/31/2010, per C. Guidry/JSC (4/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Boyd, Jennifer (Department of Psychiatry, UCSF) Marmar, Charles (Department of Psychiatry, UCSF) Weiss, Daniel (Department of Psychiatry, UCSF) Neylan, Thomas (Department of Psychiatry, UCSF)		
Grant/Contract No.:	NNJ05HG01A		
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Task Description:	In future space missions to an asteroid or Mars, crewmembers will be given more autonomy to plan and carry out their activities due to the long distances involved and the tasks that will be needed to explore a planetary surface. The impact of increased autonomy on crew performance has not been evaluated formally. In addition, while the evidence is largely anecdotal, strict adherence to timelines sent from mission control has been reported to be fatiguing and stressful for astronauts. Furthermore, such stressors can affect interpersonal interactions, resulting in increased team conflict and reduced morale and productivity. The long-term goal of our project is to conduct a scientific hypotheses-testing study involving a series of on-orbit missions to the ISS in which we measure the effects on crewmember mood, team behavior, cohesion, task performance, and the crew-mission control interaction of "high" autonomy conditions (where crewmembers plan much of their work schedule and perhaps experience a "Mars-like" communication delay with family members and mission control personnel back on Earth). This will be contrasted with "low" autonomy conditions (where mission control personnel communicate in real-time with the crew and develop the work schedule, much as happens now). The impact of high versus low crew autonomy on mission control personnel will also be studied. Before engaging in this flight experiment, we have been asked by NASA/JSC management to test the feasibility and safety of our methods during pilot studies involving a series of ground-based space analog missions (e.g., NEEMO, Houghton-Mars Project, Mars-500 105-day Simulation), where key elements of the flight experiment can be evaluated. Thus, our short-term goal will be to conduct such feasibility pilot studies involving such missions. This is the purpose of the current "Autonomy" pilot project, which deals with NEEMO and HMP simulations. (Under separate funding from NASA and the NSBRI, we have also studied the Mars-500 105-day Simulation, but this will not be reported here.)
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research addresses Critical Path Roadmap Risk No. 24 (Discipline--Behavioral Health & Performance and Space Human Factors): Human Performance Failure Due to Poor Psychosocial Adaptation. The description of this risk factor states: "Human performance failure may occur due to problems associated with adapting to the space environment, interpersonal relationships, group dynamics, team cohesiveness, and pre-mission preparation." Specific risk factors are stated as follows: "The isolated and confined nature of space flight, along with its potential hazards, pose human performance related challenges. This risk may be influenced by boredom with available food, crew autonomy and increased reliance on each other, crowding, distance from family and friends, duration of flight, incompatible crewmembers, interpersonal tensions, mechanical breakdowns, poor communications, scheduling constraints and requirements, sleep disturbances, or social isolation." Our research specifically addresses the impact of high versus low autonomy in isolated groups (such as astronauts working in space), and our measures evaluate such factors as crewmember mood, group interactions, the effects of isolation over time, crew and mission control cohesion, performance, leadership role, and the crew-ground relationship. Thus, many of the elements of this risk factor are directly explored in our research.
Task Progress:	We conducted a series of pilot studies in this project: NEEMO 12 and 13, and the Haughton-Mars Project in 2008 (HMP 2008). We were scheduled to participate in the NEEMO 14 study as well before it was cancelled. In the series of pilot studies reported below, crewmember and mission control subjects filled out a weekly Study Questionnaire, which included mood and interpersonal interactions questions from the Profile of Mood States, the Group Environment Scale, and the Work Environment Scale, as well as a critical incident log. Additionally, new measures were developed and tested in the Mars 500 105-day Simulation but not on any of the simulations described in this report. We had planned to test these in the NEEMO 14 mission, but this mission was postponed due to operational reasons unrelated to our research. Our independent variable was low versus high autonomy, and our dependent variables were scores on our measures of crewmember mood, team behavior, cohesion, task performance, and the crew-mission control interaction. It was not expected that autonomy would adversely affect mood, group interactions, and performance. In fact, it was expected that mood, group interactions, and performance would be the same or better during high autonomy periods. Results 1. NEEMO One study involved the NASA Extreme Environment Mission Operations (NEEMO) program. The purpose of this program is to study human survival in the Aquarium underwater laboratory in preparation for future space missions. The Aquarius is located near Key Largo, Florida. Missions typically last 10 to 14 days and involve research and other activities conducted by 4 or more "aquanauts", who are usually astronauts or other NASA personnel. These activities are conducted within the habitat or during simulated EVAs in the ocean. "Mission control" personnel located topside monitor crewmember progress and timeline the schedule. We studied two NEEMO missions: 12 and 13. NEEMO 12 consisted of two weeks of low autonomy, where the crewmember work schedules were timelined by the topside mission control. Subjects included 3 NASA crewmembers and 6 topside personnel. NEEMO 13 consisted of one week of low autonomy and one week of high autonomy (where the crew had more flexibility to plan their own work schedule). Subjects included 4 NASA crewmembers and 8 topside personnel. At the end of each week, subjects completed a measure of mood (Profile of Mood States), items from two measures of social climate (Group Environment Scale, Work Environment Scale), and a critical incident log. Due to the small number of data points, we could only analyze the data descriptively by examining change scores of the subscales of our measures from the first to the second week, separately for the two missions. Preliminary analyses suggested that the autonomy condition was successfully employed in NEEMO 13 with no adverse results. There seemed to be clear differences between the low and high autonomy weeks by the results, the crewmembers reported more direction from the mission commander (reflected by an increase in "Leader Control") as well as a decrease in fatigue from the low to the high autonomy weeks during NEEMO 13 as compared to NEEMO 12, which only had low autonomy conditions. For the topside personnel, work pressure was stable in NEEMO 13, whereas it rose in the second week during NEEMO 12. However, confusion increased during the high autonomy second week of NEEMO 13 as compared to the second week of NEEMO 12 (where it declined), suggesting that the topside personnel were not as clear in their roles when the crew took over more of the schedule.

2. HMP 2008

A second pilot study involved some of the Summer 2008 crewmembers working on the Haughton-Mars Project (HMP) on Devon Island in Canada. The location was an old, large impact crater which had similarities to the Martian surface in appearance and geological characteristics. People participating in this program dress up in space suits, travel in a Mars rover, and conduct tasks similar to those anticipated during an actual expedition to Mars. There is no formal “mission control”, and the participants have great freedom to plan their activities each day. Hence, there is great autonomy in the work schedule.

We studied the last three weeks of the 2008 mission. Eight non-NASA scientists and graduate students served as subjects. As in our previous work, they completed weekly POMS, GES, WES and critical incident log measures. Since we had no contrasting low autonomy period with which to compare our results, we decided to compare the subscale scores of the HMP subjects with those from published adult norms in the POMS, GES, and WES manuals, and with subscale scores from mission control subjects who participated in our previous on-orbit studies.

The HMP group reported greater levels of in-group leader direction and innovation and less need for outside direction as compared with other work groups on Earth, which presumably had more managerial control. These findings suggest that they were working under fairly high autonomous conditions. In addition, they reported being more cohesive than as compared with other work groups as well. They also reported being more tired than mission control groups, but this might have been an artifact of their working in the Canadian north in the summertime, when there is constant daylight and the work if more physically taxing than sitting in front of a mission control console. Taken together, these results suggest that people working under relatively high autonomous conditions are not adversely affected and may even bond together more cohesively than other work groups.

The results of these pilot studies suggest that in isolated and confined conditions that are meant to simulate those encountered during space missions, high autonomy is safe, well-received by crewmembers, and offers some potential advantages over low autonomy conditions. The crews have reported that they enjoy the high autonomy condition, and there have been no untoward effects to date. There have been some suggestions that high autonomy improves mood and cohesion, with more innovation and reliance on the crew leader. Outside mission monitors reported less work pressure but some confusion concerning their role. Clearly, these results are suggestive based on limited pilot data, but they point to the safety and possible advantages of high autonomy on crew functioning during actual space missions.

As can be seen, our studies directly address the elements of Critical Path Roadmap Risk No. 24, especially with reference to the psychosocial impact of high autonomy. The next step will be to complete our study of the effects of high versus low autonomy during the Mars 500 105-day Simulation and compare our findings with those from NEEMO and HMP 2008.

Together with colleagues from ESA and Russia, we have prepared and submitted a proposal to ESA for a hypotheses-testing study on autonomy to involve a series of missions to the International Space Station. If approved by ESA, then we plan to ask NASA to support our lab in conducting this study.

Bibliography Type:	Description: (Last Updated: 03/17/2017)
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